

Motivation

Natural disasters such as floods, earthquakes, and building collapses often create environments where GPS signals are unavailable or unreliable. Manual search operations are slow, dangerous, and resource-intensive. Autonomous drone systems can significantly reduce rescue time and improve survival rates by enabling rapid area coverage and real-time victim detection.

Problem Statement:

How can drones autonomously detect humans and report their location in indoor or GPS-denied environments during rescue operations?

Inspiration: Honeybee Swarm Intelligence

The system architecture is inspired by the cooperative behavior of honeybee colonies. In a natural hive, worker bees search for resources and communicate findings back to the queen. Similarly, this project uses multiple drones as "Worker" units and a Ground Control Station (GCS) as the "Queen" node.

Biological Analogy:

- Queen Bee → Ground Control Station (Command Center)
- Worker Bees → Autonomous Rescue Drones
- Hive → Search and Rescue Environment

Project Overview

This project implements a bio-inspired drone swarm system for search and rescue in GPS-denied indoor environments. Each drone is equipped with a Pixhawk flight controller, Raspberry Pi companion computer, and a camera. The drone autonomously navigates using onboard sensors and detects humans using a lightweight deep learning model. Upon detection, the drone transmits estimated coordinates to the Ground Control Station via radio telemetry, enabling rapid rescue response.



Figure 1: One Worker drone

Core Algorithms

Autonomous Navigation: SLAM (Simultaneous Localization and Mapping)

- Enables navigation in GPS-denied indoor environments
- Uses camera and IMU based Visual Odometry (VO) data to estimate drone position
- Builds a map of the environment while tracking motion

Human Detection: YOLOv8 Nano with Fuzzy Augmentation

- Lightweight deep learning model optimized for Raspberry Pi
- Detects human presence in real time
- Fuzzy augmentation improves robustness under challenging conditions

Visual Odometry Techniques:

- Uses camera with a lens mounted IMU
- IMU tracks acceleration in 3 axes
- Relative position to the launch point is determined.

Hardware and System Integration

Flight Controller:

Pixhawk manages low-level stabilization, PID control, and flight dynamics.

Companion Computer:

Raspberry Pi performs human detection, coordinate estimation, and telemetry communication.

Sensors:

- Camera for visual monitoring and detection
- MPU6050 IMU for motion sensing and position estimation

Communication:

Radio telemetry enables real-time data transmission between drone and Ground Control Station.

System Workflow and Communication

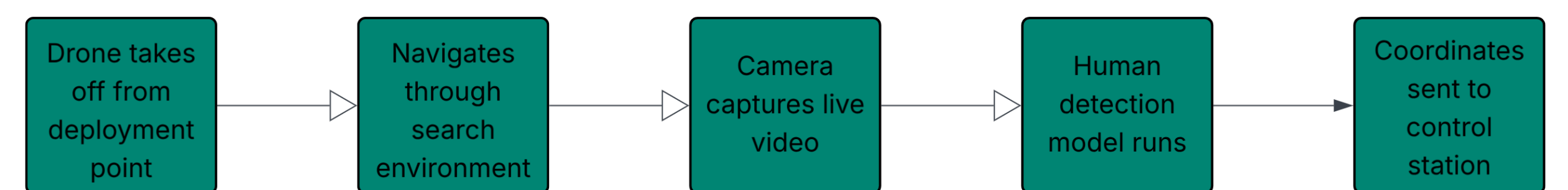


Figure 2: Flowchart of operational workflow

Results Achieved

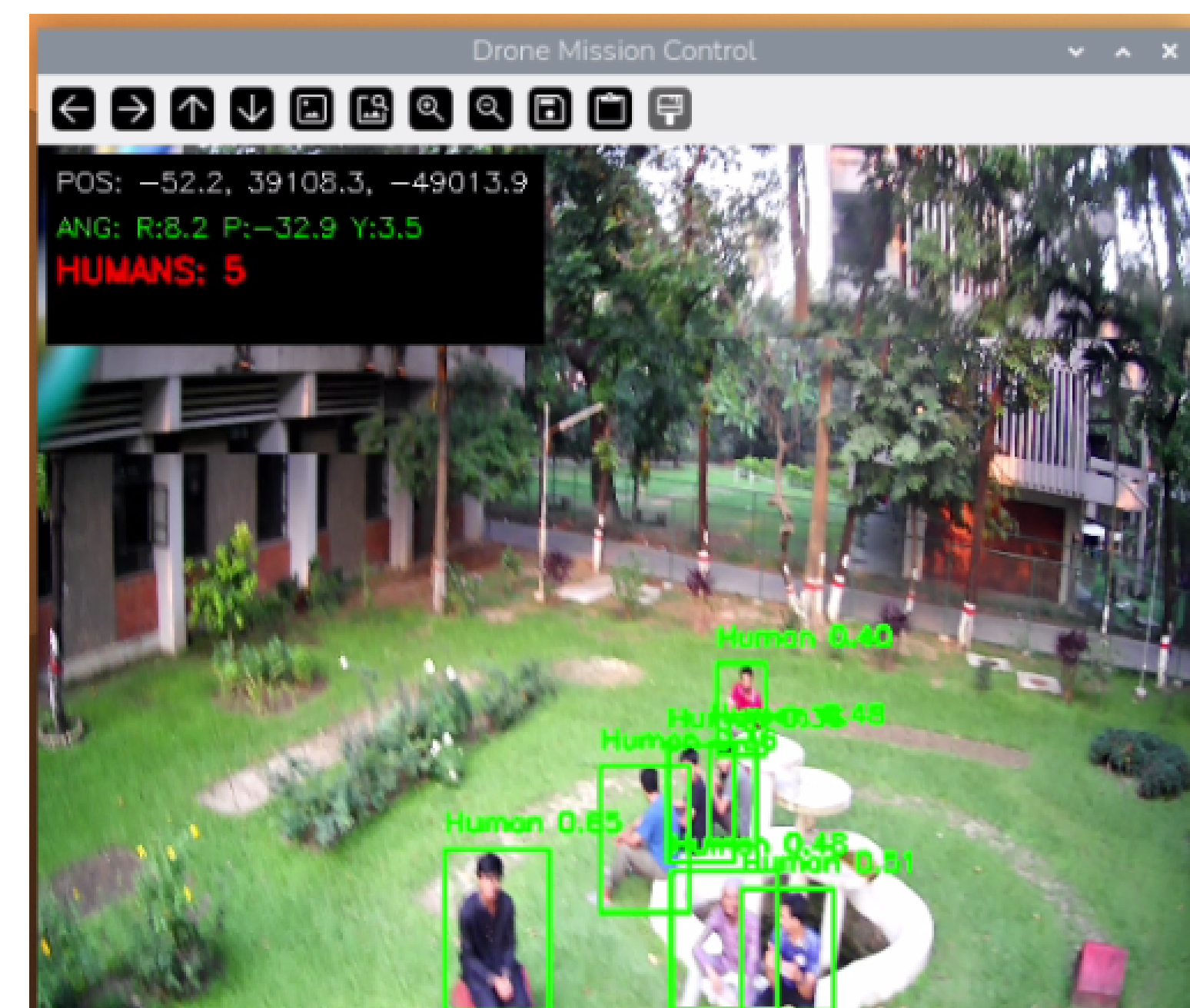


Figure 3: Mid-flight human detection

Applications

- Search and Rescue Operations
- Disaster Response and Emergency Management
- Indoor Surveillance in GPS-Denied Areas
- Flood and Building Collapse Monitoring
- Swarm-Based Autonomous Exploration

Future Work and System Expansion

Swarm Intelligence Improvements

- Fully autonomous multi-drone coordination
- Distributed swarm decision-making

Navigation Enhancements

- Advanced SLAM optimization
- Obstacle avoidance sensors
- Improved indoor localization accuracy

AI and Detection Improvements

- Thermal human detection integration
- Real-time victim classification
- Faster onboard inference models